

THE HOLTZ–STURMFELS IDEAL-GENERATION CONJECTURE FAILS FOR $n \geq 5$

ABSTRACT. Let P_n be the homogeneous prime ideal of the projective variety of principal minors of symmetric $n \times n$ matrices, and let J_n be the ideal generated by the hyperdeterminantal module $\text{HD}(n)$. Holtz and Sturmfels conjectured that $J_n = P_n$. We work over $\mathbb{C}$. For $n = 5$, we construct five SL_2^5 -invariant quartics $B_1, \dots, B_5$. Their pullbacks under the principal-minor map are all $12t^{10}\Pi^2$, where Π is the pentad. Hence the four differences $B_1 - B_i$, $2 \leq i \leq 5$, lie in $(P_5)_4$; a 4×4 coefficient determinant equal to -512 shows that they are linearly independent. The module $\text{HD}(5)$ has dimension 250 and contains no nonzero SL_2^5 -invariant, so $\dim(P_5)_4 \geq 254$ and $J_5 \neq P_5$. The same relations on any five principal indices show that $J_n \neq P_n$ for every $n \geq 5$. Since Oeding proved $\sqrt{J_n} = P_n$, the hyperdeterminantal ideal is nonradical for every $n \geq 5$. All polynomials exhibited have integer coefficients, and the real form of the conjecture — the form in which it was posed — fails for every $n \geq 5$ as well.

1. STATEMENT OF THE RESULT

We work over $\mathbb{C}$ throughout, as [4] does; [1] pose Conjecture 14 over $\mathbb{R}$, and the Status paragraph at the end records the descent to that form.

For $V_i = \mathbb{C}^2$, set

$$V^{(n)} = V_1 \otimes \cdots \otimes V_n, \quad R_n = \text{Sym}((V^{(n)})^*) = \mathbb{C}[A_I : I \subseteq [n]].$$

Let X_I denote the principal submatrix of a symmetric matrix X indexed by I . The homogeneous principal-minor map is

$$\phi_n : \mathbb{P}(\text{Sym}^2 \mathbb{C}^n \oplus \mathbb{C}) \dashrightarrow \mathbb{P}(V^{(n)}), \quad [X : t] \mapsto [t^{n-|I|} \det X_I]_{I \subseteq [n]}.$$

Let Z_n be the closure of its image and let $P_n = I(Z_n)$. Let

$$G_n = \text{SL}(V_1) \times \cdots \times \text{SL}(V_n)$$

act on R_n through $V^{(n)}$, and let the symmetric group S_n act on R_n by permuting the n tensor factors, hence the indices of the coordinates A_I . Write $\text{HD}(n)$ for the *hyperdeterminantal module*: the linear span of the orbit of Cayley’s $2 \times 2 \times 2$ hyperdeterminant under $G_n \rtimes S_n$. This is the module of [1, §6] and the module HD of [4, §5]. Equivalently, $\text{HD}(n)$ is the G_n -module generated by the hyperdeterminants of all $\binom{n}{3} 2^{n-3}$ two-by-two-by-two subtensors of $(A_I)_{I \subseteq [n]}$: by (5) and its analogue for general n

2020 *Mathematics Subject Classification*. 13A50, 14M12, 15A15, 15A69.

Key words and phrases. principal minors, symmetric matrix, hyperdeterminant, invariant theory.

in the proof of Theorem 1, the summand attached to a triple of tensor directions is an irreducible G_n -module and contains the hyperdeterminants of all 2^{n-3} subtensors that use those three directions, while S_n permutes the $\binom{n}{3}$ triples. Both groups are needed: the G_n -orbit span of a single subtensor hyperdeterminant is one summand of (5), of dimension 25 rather than 250 when $n = 5$. One has $\text{HD}(n) \subseteq (P_n)_4$ [4, Proposition 5.5], and we set

$$J_n = (\text{HD}(n)) \subseteq R_n.$$

Holtz and Sturmfels conjectured that $J_n = P_n$ [1, Conjecture 14]. Oeding proved the set-theoretic equality $\mathcal{V}(J_n) = Z_n$ [4, Theorem 1.3]; the ideal-theoretic problem was subsequently recorded as open for $n \geq 5$ [2, p. 1194].

We use Oeding's representation-theoretic description of $\text{HD}(n)$. In particular,

$$(1) \quad \dim \text{HD}(n) = \binom{n}{3} 5^{n-3} \quad [4, \text{Observation 5.1}].$$

The formula printed for the dimension of the same module in [1, §6] is $\binom{2^{n-3}+3}{4} \binom{n}{3}$. The two agree for $n \leq 4$, giving 1 and 20, and disagree from $n = 5$ on, where they give 250 and 350. The disagreement is one of counts and not of modules: both papers define $\text{HD}(n)$ as the span of the orbit of the $2 \times 2 \times 2$ hyperdeterminant under G_n and S_n , and Conjecture 14 is the assertion that this span generates P_n , so that span is the object of Theorem 1 below. Neither value is used in the refutation. What separates our quartics from $\text{HD}(n)$ in Proposition 5 and in the proof of Theorem 1 is the G_n -isotypic type of $\text{HD}(n)$, which forces $\text{HD}(n)^{G_n} = 0$ whatever the dimension of $\text{HD}(n)$ may be; (1) enters only in the numerical bound printed in Theorem 1.

Theorem 1. *For every $n \geq 5$,*

$$J_n \subsetneq P_n.$$

More precisely, $(P_5)_4$ contains a four-dimensional SL_2^5 -trivial subspace W with $W \cap \text{HD}(5) = 0$, and hence

$$\dim(P_5)_4 \geq 254 > 250 = \dim \text{HD}(5).$$

Consequently, J_n is nonradical for every $n \geq 5$.

2. FIVE INVARIANT QUARTICS

Identify $I \subseteq [5]$ with its indicator $\alpha \in \{0, 1\}^5$, and write $A_\alpha = A_I$. Let ε be the alternating form on $\mathbb{C}^2$, normalized by

$$\varepsilon_{01} = 1, \quad \varepsilon_{10} = -1, \quad \varepsilon_{00} = \varepsilon_{11} = 0.$$

For $i \in [5]$, define

$$(2) \quad q_{ab}^{(i)}(A) = \sum_{\substack{\alpha, \beta \in \{0, 1\}^5 \\ \alpha_i = a, \beta_i = b}} A_\alpha A_\beta \prod_{j \neq i} \varepsilon_{\alpha_j \beta_j}.$$

Set

$$Q_i(A) = (q_{ab}^{(i)}(A))_{a,b=0}^1, \quad B_i(A) = \det Q_i(A).$$

The four contractions make $Q_i(A)$ symmetric. This is the standard construction of the five degree-four invariants of a five-qubit tensor; see [3].

Lemma 2. *Each B_i is invariant under G_5 .*

Proof. The alternating form is SL_2 -invariant. Contracting the four factors other than i therefore gives

$$Q_i(g \cdot A) = g_i Q_i(A) g_i^\top \quad (g = (g_1, \dots, g_5) \in G_5),$$

where g_i acts through the identification of V_i with V_i^* furnished by ε ; without that identification the right-hand side reads $g_i^{-\top} Q_i(A) g_i^{-1}$. Taking determinants and using $\det g_i = 1$, which disposes of either form, proves the claim. $\square$

3. THE QUARTIC RELATIONS

For a symmetric 5×5 matrix $X = (x_{ij})$, define the pentad

$$\begin{aligned} \Pi(X) = & x_{12}x_{13}x_{24}x_{35}x_{45} - x_{12}x_{13}x_{25}x_{34}x_{45} - x_{12}x_{14}x_{23}x_{35}x_{45} \\ & + x_{12}x_{14}x_{25}x_{34}x_{35} + x_{12}x_{15}x_{23}x_{34}x_{45} - x_{12}x_{15}x_{24}x_{34}x_{35} \\ & + x_{13}x_{14}x_{23}x_{25}x_{45} - x_{13}x_{14}x_{24}x_{25}x_{35} - x_{13}x_{15}x_{23}x_{24}x_{45} \\ & + x_{13}x_{15}x_{24}x_{25}x_{34} + x_{14}x_{15}x_{23}x_{24}x_{35} - x_{14}x_{15}x_{23}x_{25}x_{34}. \end{aligned}$$

Proposition 3. *For every $i \in [5]$,*

$$(3) \quad B_i(\phi_5(X, t)) = 12t^{10}\Pi(X)^2$$

as an identity in $\mathbb{Z}[x_{ij}, t : 1 \leq i \leq j \leq 5]$.

Proof. If a summand in (2) is nonzero, then $\alpha_j + \beta_j = 1$ for every $j \neq i$. Thus $q_{ab}^{(i)}$ has subset weight $4 + a + b$, and every monomial of B_i has subset weight 10. It is therefore enough to verify (3) on the affine chart $t = 1$.

Substituting the thirty-two principal minors $A_I = \det X_I$ into (2) and expanding over $\mathbb{Z}$ gives

$$B_i((\det X_I)_{I \subseteq [5]}) = 12\Pi(X)^2 \quad (i = 1, \dots, 5).$$

This is an exact polynomial calculation, and it is specified completely by what precedes: each side has exactly seventy-three monomials, all of degree ten in the fifteen entries x_{jk} , and each of the five differences $B_i((\det X_I)_{I \subseteq [5]}) - 12\Pi(X)^2$ vanishes coefficientwise. No step of the expansion is cancellation-sensitive, and it is reproduced in exact integer arithmetic, with no sampling and no floating point, by any computer algebra system. $\square$

Set

$$F_i = B_1 - B_i, \quad i = 2, 3, 4, 5.$$

Proposition 3 implies

$$(4) \quad F_2, F_3, F_4, F_5 \in (P_5)_4.$$

Lemma 4. *The quartics F_2, F_3, F_4, F_5 are linearly independent.*

Proof. Consider the coefficients of

$$\begin{aligned} m_1 &= A_{134}A_{234}A_{15}A_{25}, & m_2 &= A_{124}A_{234}A_{15}A_{35}, \\ m_3 &= A_{124}A_{134}A_{25}A_{35}, & m_4 &= A_{123}A_{234}A_{15}A_{45}. \end{aligned}$$

With columns ordered as (F_2, F_3, F_4, F_5) , the coefficient matrix is

$$\begin{pmatrix} 0 & 4 & 4 & 4 \\ 4 & 0 & 4 & 4 \\ -4 & -4 & 0 & 0 \\ 4 & 4 & 0 & 4 \end{pmatrix}, \quad \det = -512.$$

Hence the four quartics are independent. $\square$

4. SEPARATION FROM THE HYPERDETERMINANTAL MODULE

As a G_5 -module, $\text{HD}(5)$ decomposes as

$$(5) \quad \text{HD}(5) \cong \bigoplus_{\substack{T \subseteq [5] \\ |T|=3}} \left(\bigotimes_{j \in T} \mathbf{S}_{(2,2)} V_j^* \right) \otimes \left(\bigotimes_{j \notin T} \mathbf{S}_{(4)} V_j^* \right).$$

This is Oeding's compact module notation, defined at [4, p. 85] as the sum over those permutations of a partition tuple that yield distinct G_n -modules, and restated inside the proof of [4, Proposition 5.2]; that proposition shows in addition that HD occurs with multiplicity one in $\text{Sym}^4((V^{(n)})^*)$ and is irreducible as a $G_n \rtimes S_n$ -module. The dimension (1) is the count of exactly this direct sum [4, Observation 5.1].

Now $\mathbf{S}_{(2,2)}\mathbb{C}^2$ is one-dimensional and trivial on SL_2 , whereas $\mathbf{S}_{(4)}\mathbb{C}^2 \cong \text{Sym}^4(\mathbb{C}^2)$ has dimension five and no nonzero SL_2 -fixed vector. Therefore

$$(6) \quad \dim \text{HD}(5) = \binom{5}{3} 5^2 = 250, \quad \text{HD}(5)^{G_5} = 0.$$

Proposition 5. *The degree-four component $(P_5)_4$ strictly contains $\text{HD}(5)$. Indeed,*

$$W = \text{span}\{F_2, F_3, F_4, F_5\}$$

is a four-dimensional G_5 -trivial subspace of $(P_5)_4$ and $W \cap \text{HD}(5) = 0$.

Proof. Membership and $\dim W = 4$ follow from (4) and Lemma 4. Lemma 2 shows that G_5 fixes W pointwise, while (6) gives $W \cap \text{HD}(5) = 0$. Since $\text{HD}(5) \subseteq (P_5)_4$ by [4, Proposition 5.5],

$$\dim(P_5)_4 \geq \dim \text{HD}(5) + \dim W = 254.$$

Finally, $(J_5)_4 = \text{HD}(5)$, because J_5 is generated in degree four. Hence $J_5 \neq P_5$. $\square$

Proof of Theorem 1. Proposition 5 proves the assertion for $n = 5$. Let $n > 5$, fix a five-element set $S \subseteq [n]$, and identify the coordinates A_I , $I \subseteq S$, with the coordinates of R_5 . Let $F_i^S \in R_n$ be the resulting copy of F_i . For every symmetric $n \times n$ matrix X , the subvector of $\phi_n(X, t)$ indexed by $I \subseteq S$ is $t^{n-5}\phi_5(X_S, t)$. Since F_i is homogeneous of degree four,

$$F_i^S(\phi_n(X, t)) = t^{4(n-5)}F_i(\phi_5(X_S, t)) = 0.$$

Thus $F_i^S \in (P_n)_4$.

Set $H_S = \prod_{j \in S} \mathrm{SL}(V_j)$. Each F_i^S is fixed by H_S . On the other hand, restriction of Oeding’s decomposition gives

$$\mathrm{HD}(n)|_{\prod_{j=1}^n \mathrm{SL}(V_j)} \cong \bigoplus_{\substack{T \subseteq [n] \\ |T|=3}} \left(\bigotimes_{j \in T} \mathbf{S}_{(2,2)} V_j^* \right) \otimes \left(\bigotimes_{j \notin T} \mathbf{S}_{(4)} V_j^* \right).$$

For every T with $|T| = 3$, at least two indices of S lie outside T . Thus every summand contains a nontrivial $\mathrm{Sym}^4(V_j^*)$ factor for the action of H_S , and consequently

$$\mathrm{HD}(n)^{H_S} = 0.$$

Since $F_i^S \neq 0$, it follows that $F_i^S \notin \mathrm{HD}(n) = (J_n)_4$. Therefore $J_n \neq P_n$ for every $n \geq 5$.

Finally, $\mathcal{V}(J_n) = Z_n$ [4, Theorem 1.3] and the Nullstellensatz give $\sqrt{J_n} = I(Z_n) = P_n$. Hence J_n is nonradical for every $n \geq 5$. $\square$

Exact verification. Every check described here is a recomputation from the data printed in Sections 1–4, in exact integer arithmetic and with no floating point, so a referee can redo it from this paper alone. The computation has been re-run once, independently, on a separate machine, by a program written from those sections alone, and all 36 of its checks agree with what is reported here. It re-derives the pentad as the signed sum over the twelve Hamiltonian cycles of K_5 and recovers the twelve monomials printed in Section 3; it verifies (3) coefficientwise in $\mathbb{Z}[x_{jk}, t]$ for all five i , seventy-three monomials on each side and no nonzero residual; it recomputes the matrix of Lemma 4, its determinant -512 , and the rank 4 of the full 4×280 coefficient matrix of $F_2, \dots, F_5$; it obtains $\dim \mathbf{S}_{(2,2)} \mathbb{C}^2 = 1$, $\dim \mathbf{S}_{(4)} \mathbb{C}^2 = 5$ and $\mathrm{Sym}^m(\mathbb{C}^2)^{\mathrm{SL}_2} = 0$ for $m > 0$, hence (6); it builds the torus-weight-zero part of $\mathrm{HD}(5)$ explicitly, finds its rank to be 10 and finds that adjoining $F_2, \dots, F_5$ raises the rank to 14, which confirms $W \cap \mathrm{HD}(5) = 0$ by computation as well as by (6) — a confirmation only, since the matching upper bound $\dim \mathrm{HD}(5)^{\mathrm{wt} 0} \leq 10$ was there obtained by sampling the orbit, whereas the proof of Proposition 5 uses (6) and not this computation; it checks that Cayley’s hyperdeterminant of each of the forty $2 \times 2 \times 2$ subtensors of $\phi_5(X, t)$ is the zero polynomial; and it re-runs the reduction in the proof of Theorem 1 for $n = 6, \dots, 12$ over every one of the 1715 five-element subsets and

every i , together with the module facts for $n = 3, \dots, 14$ over every pair of a three-element and a five-element subset.

Two limits of that re-run should be recorded. Theorem 1 asserts $J_n \neq P_n$ for every $n \geq 5$, an infinite family: the re-run covers $n = 5$ exactly and symbolically, the relations for $n = 6, \dots, 12$ at exact integer points, and the module facts for $n \leq 14$, but not the relations for $n = 13, 14$ and nothing at all for $n > 14$; those rest on the uniform argument in the proof of Theorem 1 and not on computation. And the inclusion $\text{HD}(n) \subseteq (P_n)_4$ was re-derived only at the orbit generators for $n = 5$, by the vanishing check just described, while $\sqrt{J_n} = P_n$ was taken as cited and not re-derived. The program and its transcript accompany this note. The claim recorded here is that an independent re-run agrees with what is reported above, not that it constitutes a second proof.

Status and scope. We have worked over $\mathbb{C}$, as [4] does, whereas Conjecture 14 is posed in [1] over $\mathbb{R}$, with ring $\mathbb{R}[A_\bullet]$ and group $\text{SL}_2(\mathbb{R})^n$. The conclusion descends to that form. The quartics F_i and F_i^S have integer coefficients and (3) is an identity over $\mathbb{Z}$, so each is a relation among the principal minors of a real symmetric matrix. And $\text{HD}(n)$, the span of the orbit of a quartic with integer coefficients under a group defined over $\mathbb{Q}$, has a basis of quartics with integer coefficients, so a nonzero real element of the intersection of $\text{HD}(n)$ with the real span of the F_i^S would be a nonzero element of the complex intersection, which is 0. Hence the real form of Conjecture 14 fails for every $n \geq 5$ as well.

What is refuted is the ideal-theoretic statement, and only for $n \geq 5$. Conjecture 14 holds for $n = 3$, where P_3 is principal, and for $n = 4$, where it is verified by the computations of [1, §4]; nothing here touches those cases. No minimality is claimed: $\dim(P_5)_4 \geq 254$ is an inequality, we do not determine $\dim(P_5)_4$, and we exhibit no generating set for P_n . Oeding's set-theoretic theorem is untouched — $\mathcal{V}(J_n) = Z_n$ still holds — and what fails is the equality of ideals.

The five quartics $B_1, \dots, B_5$ are not new: they are the five degree-four invariants of a five-qubit state constructed by Luque and Thibon [3], obtained by fixing one qubit, taking the degree-two invariant of the remaining quadri-linear form, and then its discriminant. What is new here is the common value $12t^{10}\Pi^2$ of their pullbacks (3), and with it the membership (4). We have not found the strict inclusion $\text{HD}(5) \subsetneq (P_5)_4$, or the nonradicality of J_n , recorded anywhere: the ideal-theoretic problem is stated as open in [2, p. 1194], and the literature citing [1] and [4] that we were able to examine treats the variety of principal minors of symmetric matrices set-theoretically. That is an absence in a search we cannot claim to be exhaustive, and not a proof of absence.

REFERENCES

- [1] O. Holtz and B. Sturmfels, *Hyperdeterminantal relations among symmetric principal minors*, J. Algebra **316** (2007), no. 2, 634–648, doi:10.1016/j.jalgebra.2007.01.039.
- [2] H. Huang and L. Oeding, *Symmetrization of principal minors and cycle-sums*, Linear Multilinear Algebra **65** (2017), no. 6, 1194–1219, doi:10.1080/03081087.2016.1233932.
- [3] J.-G. Luque and J.-Y. Thibon, *Algebraic invariants of five qubits*, J. Phys. A **39** (2006), no. 2, 371–377, doi:10.1088/0305-4470/39/2/007.
- [4] L. Oeding, *Set-theoretic defining equations of the variety of principal minors of symmetric matrices*, Algebra Number Theory **5** (2011), no. 1, 75–109, doi:10.2140/ant.2011.5.75.